

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product name : THYPAAN ZN

Manufacturer or supplier's details

Company : FMC Agro Kazakhstan LLP

Address : Timiryazev str. 28V, office 501, 5 floor
050040 Almaty
Kazakhstan

Emergency telephone number : +44 20 3885 0382 (CHEMTREC's European Regional Toll-Free Number)
1 703 / 741-5970 (CHEMTREC - International)
1 703 / 527-3887 (CHEMTREC - Alternate)

Medical Emergency Number : Ambulance in the Republic of Kazakhstan: 103 (24/7, toll-free).
Almaty City Clinical Hospital, Toxicology Department, 96 Kazybek Bi Street: +7 727 317 01 69 – hotline; +7 727 292 31 01 – reception.
+1 651 / 632-6793 (outside USA & Canada, collect call)

E-mail address : SDS-Info@fmc.com

Recommended use of the chemical and restrictions on use

Recommended use : Crop nutrition

Restrictions on use : Use as recommended by the label.

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Dermal) : Category 5

Skin irritation : Category 2

Reproductive toxicity : Category 2

Specific target organ toxicity - repeated exposure (Oral) : Category 2 (Central nervous system, Reproductive organs)

Short-term (acute) aquatic hazard : Category 1

Long-term (chronic) aquatic hazard : Category 1

GHS-Labeling

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0 Revision Date: 26.11.2025 SDS Number: 50001138 Date of last issue: -
Date of first issue: 26.11.2025

Hazard pictograms :   

Signal word : Warning

Hazard statements : H313 May be harmful in contact with skin.
H315 Causes skin irritation.
H361 Suspected of damaging fertility or the unborn child.
H373 May cause damage to organs (Central nervous system, Reproductive organs) through prolonged or repeated exposure if swallowed.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements : **Prevention:**
P201 Obtain special instructions before use.
P260 Do not breathe mist or vapours.
P264 Wash skin thoroughly after handling.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.
Disposal:
P501 Dispose of contents/ container to an approved facility in accordance with local, regional, national and international regulations.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Components

Chemical name	CAS-No.	Classification	MAC value mg/m ³ / TSEL value	Concentration (% w/w)
zinc oxide	1314-13-2	Acute Tox.5; H303 Acute Tox.5; H313 Aquatic Acute1; H400 Aquatic Chronic1; H410	MPC-TWA: 0.5 mg/m ³ Class 2 - Highly dangerous Data Source: RU OEL MPC-STEL: 1.5 mg/m ³ Class 2 - Highly dangerous Data Source: RU OEL	>= 30 - < 50

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0 Revision Date: 26.11.2025 SDS Number: 50001138 Date of last issue: -
Date of first issue: 26.11.2025

ethanediol	107-21-1	Acute Tox.5; H313 STOT RE2; H373 (Kidney) Aquatic Acute3; H402	MPC-TWA: 5 mg/m ³ Class 3 - Moder- ately dangerous Data Source: RU OEL MPC-STEL: 10 mg/m ³ Class 3 - Moder- ately dangerous Data Source: RU OEL	>= 2.5 - < 10
1,2-benzisothiazol-3(2H)-one	2634-33-5	Acute Tox.4; H302 Eye Dam.1; H318 Skin Sens.1; H317 Aquatic Acute1; H400 Aquatic Chronic2; H411	No data available	>= 0.0025 - < 0.025

For explanation of abbreviations see section 16.

4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Show this safety data sheet to the doctor in attendance.
Do not leave the victim unattended.
- If inhaled : Move to fresh air.
If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.
- In case of skin contact : Take off all contaminated clothing immediately.
Wash contaminated clothing before re-use.
Wash off immediately with plenty of water for at least 15 minutes.
Get medical attention immediately if irritation develops and persists.
- In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Do not induce vomiting without medical advice.
Keep respiratory tract clear.
Do not give milk or alcoholic beverages.

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.

Most important symptoms and effects, both acute and delayed : May be harmful in contact with skin.
Causes skin irritation.
Suspected of damaging fertility or the unborn child.
May cause damage to organs through prolonged or repeated exposure if swallowed.

Protection of first-aiders : First Aid responders should pay attention to self-protection and use the recommended protective clothing
Avoid inhalation, ingestion and contact with skin and eyes.
If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.

Unsuitable extinguishing media : Do not spread spilled material with high-pressure water streams.

Specific hazards during fire-fighting : Do not allow run-off from fire fighting to enter drains or water courses.

Hazardous combustion products : Fire may produce irritating, corrosive and/or toxic gases.
Carbon oxides
metal fumes

Specific extinguishing methods : Remove undamaged containers from fire area if it is safe to do so.
Use a water spray to cool fully closed containers.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

Special protective equipment for firefighters : Firefighters should wear protective clothing and self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Evacuate personnel to safe areas.
Use personal protective equipment.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
Never return spills in original containers for re-use.
Mark the contaminated area with signs and prevent access to unauthorized personnel.

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

Only qualified personnel equipped with suitable protective equipment may intervene.
For disposal considerations see section 13.

Environmental precautions : Prevent further leakage or spillage if safe to do so.
Prevent product from entering drains.
If the product contaminates rivers and lakes or drains inform respective authorities.

Methods and materials for containment and cleaning up : Never return spills in original containers for re-use.
Collect as much of the spill as possible with a suitable absorbent material.
Pick up and transfer to properly labelled containers.
Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

Advice on protection against fire and explosion : Normal measures for preventive fire protection.

Advice on safe handling : Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Dispose of rinse water in accordance with local and national regulations.

Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.

Recommended storage temperature : > 5 °C

Further information on storage stability : No decomposition if stored and applied as directed.
Do not freeze.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Personal protective equipment

Respiratory protection : In the case of dust or aerosol formation use respirator with an approved filter.

Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate,

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

butyl rubber or nitrile rubber.

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles
Face-shield

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures : Plan first aid action before beginning work with this product.

Hygiene measures : Avoid contact with skin, eyes and clothing.
Do not inhale aerosol.
When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state : liquid

Form : suspension

Colour : white

Odour : Faint odour

Odour Threshold : No data available

pH : 8.0 - 12.0
Concentration: 100 %

Melting point/ range : No data available

Boiling point/boiling range : No data available

Flash point : No data available

Evaporation rate : No data available

Self-ignition : No data available

Upper explosion limit / Upper flammability limit : No data available

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

Lower explosion limit / Lower flammability limit	:	No data available
Vapour pressure	:	No data available
Relative vapour density	:	No data available
Relative density	:	No data available
Density	:	No data available
Bulk density	:	No data available
Solubility(ies)		
Water solubility	:	No data available
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Auto-ignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity		
Viscosity, dynamic	:	No data available
Viscosity, kinematic	:	No data available
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing
Molecular weight	:	Not applicable
Particle size	:	No data available

10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions	:	No decomposition if stored and applied as directed.
Conditions to avoid	:	Avoid extreme temperatures Avoid formation of aerosol. Protect from frost, heat and sunlight.

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

Incompatible materials : Avoid strong acids, bases, and oxidizers

Hazardous decomposition products : Toxic fumes

11. TOXICOLOGICAL INFORMATION

Acute toxicity

May be harmful in contact with skin.

Product:

Acute oral toxicity : Acute toxicity estimate: > 5,000 mg/kg
Method: Calculation method

Acute inhalation toxicity : Acute toxicity estimate: > 20 mg/l
Exposure time: 4 h
Test atmosphere: vapour
Method: Calculation method

Acute dermal toxicity : Acute toxicity estimate: 4,865 mg/kg
Method: Calculation method

Components:

zinc oxide:

Acute oral toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 423

LD50 (Mouse, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 401
Target Organs: Liver, Heart, spleen, Stomach, Pancreas
Symptoms: Damage
Remarks: mortality

Acute inhalation toxicity : LC50 (Rat, male and female): > 1.79 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: EPA OPP 81 - 3
Remarks: no mortality

Acute dermal toxicity : LD50 Dermal (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 402

ethanediol:

Acute inhalation toxicity : LC50 (Rat, male and female): > 2.5 mg/l
Exposure time: 6 h
Test atmosphere: dust/mist
Remarks: no mortality

Acute dermal toxicity : LD50 (Mouse, male and female): > 3,500 mg/kg

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

1,2-benzisothiazol-3(2H)-one:

Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Causes skin irritation.

Product:

Remarks : No data available

Components:

zinc oxide:

Species : reconstructed human epidermis (RhE)
Method : OECD Test Guideline 431
Result : No skin irritation

ethanediol:

Species : Rabbit
Result : No skin irritation

1,2-benzisothiazol-3(2H)-one:

Species : Rabbit
Exposure time : 72 h
Method : OECD Test Guideline 404
Result : No skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Remarks : Vapours may cause irritation to the eyes, respiratory system and the skin.

Components:

zinc oxide:

Species : Rabbit
Result : No eye irritation
Method : OECD Test Guideline 405

ethanediol:

Species : Rabbit
Result : No eye irritation

1,2-benzisothiazol-3(2H)-one:

Species : Bovine cornea
Result : No eye irritation

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

Method	:	OECD Test Guideline 437
Species	:	Rabbit
Result	:	Irreversible effects on the eye
Method	:	EPA OPP 81-4

Respiratory or skin sensitisation

Skin sensitisation

Based on available data, the classification criteria are not met.

Respiratory sensitisation

Based on available data, the classification criteria are not met.

Product:

Remarks	:	No data available
---------	---	-------------------

Components:

zinc oxide:

Test Type	:	Maximisation Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Does not cause skin sensitisation.
Test Type	:	Maximisation Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Substance is not considered to be potential skin sensitiser.

ethanediol:

Test Type	:	Maximisation Test
Species	:	Guinea pig
Result	:	Does not cause skin sensitisation.

1,2-benzisothiazol-3(2H)-one:

Test Type	:	Maximisation Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	May cause sensitisation by skin contact.
Species	:	Guinea pig
Method	:	FIFRA 81.06
Result	:	May cause sensitisation by skin contact.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:

zinc oxide:

Genotoxicity in vitro	:	Test Type: reverse mutation assay Method: Mutagenicity (Salmonella typhimurium - reverse mu-
-----------------------	---	---

THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

tation assay)
Result: negative

Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: equivocal

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster fibroblasts
Method: OECD Test Guideline 473
Result: negative

Test Type: Chromosome aberration test in vitro
Test system: Human lymphocytes
Result: positive

Test Type: Micronucleus test
Test system: Human epithelioid cells
Method: OECD Test Guideline 487
Result: negative

Test Type: Micronucleus test
Test system: Human lymphocytes
Result: positive

Genotoxicity in vivo : Test Type: In vivo micronucleus test
Species: Mouse (male)
Application Route: Intraperitoneal injection
Method: OECD Test Guideline 474
Result: negative

ethanediol:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OPPTS 870.5100
Result: negative

Genotoxicity in vivo : Test Type: dominant lethal test
Species: Rat
Application Route: Oral
Result: negative

1,2-benzisothiazol-3(2H)-one:

Genotoxicity in vitro : Test Type: gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative

Test Type: Ames test
Method: OECD Test Guideline 471
Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

Result: positive

Genotoxicity in vivo : Test Type: unscheduled DNA synthesis assay
Species: Rat (male)
Cell type: Liver cells
Application Route: Ingestion
Exposure time: 4 h
Method: OECD Test Guideline 486
Result: negative

Test Type: Micronucleus test
Species: Mouse
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

zinc oxide:

Species : Mouse, male and female
Application Route : Oral
Exposure time : 1 year
Dose : 4400, 22000 mg/l
NOAEL : > 22,000 mg/l
Result : negative
Remarks : Based on data from similar materials

Carcinogenicity - Assessment : Animal testing did not show any carcinogenic effects.

ethanediol:

Species : Mouse
Application Route : Oral
Exposure time : 24 month(s)
Result : negative

Reproductive toxicity

Suspected of damaging fertility or the unborn child.

Components:

zinc oxide:

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Application Route: Oral
Dose: 7.5, 15, 30mg/kg bw/day
Frequency of Treatment: 7 days/week
General Toxicity - Parent: LOAEL: 7.5 mg/kg body weight

THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

General Toxicity F1: LOAEL: 30 mg/kg body weight
Method: OECD Test Guideline 416
Result: negative
Remarks: Based on data from similar materials

Test Type: one-generation reproductive toxicity
Species: Rat, male
Application Route: Oral
Dose: 4,000 milligram per liter
Frequency of Treatment: 32 daily
General Toxicity - Parent: LOAEL: 4,000 mg/l
General Toxicity F1: LOAEL: 4,000 mg/l
Symptoms: Reduced fertility
Target Organs: male reproductive organs
Result: positive
Remarks: Based on data from similar materials

Effects on foetal development : Species: Rat
Application Route: inhalation (dust/mist/fume)
Dose: .0003, 0.002, 0.008 milligram per liter
Duration of Single Treatment: 14 d
General Toxicity Maternal: LOAEC: 0.008 mg/L
Developmental Toxicity: NOAEC: 0.008 mg/L
Embryo-foetal toxicity: NOAEC Mating/Fertility: 0.008 mg/L
Method: OECD Test Guideline 414
Result: negative

1,2-benzisothiazol-3(2H)-one:

Effects on fertility : Species: Rat, male
Application Route: Ingestion
General Toxicity - Parent: NOAEL: 18.5 mg/kg body weight
General Toxicity F1: NOAEL: 48 mg/kg body weight
Fertility: NOAEL: 112 mg/kg bw/day
Symptoms: No effects on reproduction parameters
Method: OPPTS 870.3800
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

STOT - single exposure

Based on available data, the classification criteria are not met.

STOT - repeated exposure

May cause damage to organs (Central nervous system, Reproductive organs) through prolonged or repeated exposure if swallowed.

Components:**ethanediol:**

Exposure routes : Oral
Target Organs : Kidney
Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.

THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

1,2-benzisothiazol-3(2H)-one:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity**Components:****zinc oxide:**

Species : Rat, male and female
NOAEL : 31.52 mg/kg
LOAEL : 127.52 mg/kg
Application Route : Oral
Exposure time : 13 weeks
Dose : 0, 31.52, 127.52 mg/kg
Method : OECD Test Guideline 408
Target Organs : Pancreas
Symptoms : Necrosis
Remarks : Based on data from similar materials

Species : Mouse, male and female
NOEL : 3000 ppm
Application Route : Oral
Exposure time : 13 weeks
Dose : 0, 300, 3000, 30000 ppm
Method : OECD Test Guideline 408
Remarks : Based on data from similar materials

Species : Rat, male
LOAEL : 0.0045 mg/l
Application Route : inhalation (dust/mist/fume)
Exposure time : 3 months
Dose : 0.0003, 0.0015, 0.004mg/l
Method : OECD Test Guideline 413
Target Organs : Lungs
Remarks : mortality

Species : Rat, male and female
LOAEL : 75 mg/kg bw/day
Application Route : Dermal
Exposure time : 28d
Dose : 0, 75, 180, 360 mg/kg bw/day
Method : OECD Test Guideline 410

ethanediol:

Species : Rat
NOAEL : 150 mg/kg
Application Route : Oral
Exposure time : 12 Months

Species : Dog
NOAEL : > 2,200 - < 4,400 mg/kg

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

Application Route : Dermal
Exposure time : 4 Weeks
Method : OECD Test Guideline 410

1,2-benzisothiazol-3(2H)-one:

Species : Rat, male and female
NOAEL : 15 mg/kg
Application Route : Ingestion
Exposure time : 28 d
Method : OECD Test Guideline 407
Symptoms : Irritation

Species : Rat, male and female
NOAEL : 69 mg/kg
Application Route : Ingestion
Exposure time : 90 d
Symptoms : Irritation, Reduced body weight

Aspiration toxicity

Based on available data, the classification criteria are not met.

Experience with human exposure

Components:

zinc oxide:

Inhalation : Symptoms: Fatigue, Sweating, bitter taste, chills, dry mouth, flu-like symptoms

Ingestion : Symptoms: Gastrointestinal discomfort

Further information

Product:

Remarks : No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

zinc oxide:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): 1.55 mg/l
Exposure time: 96 h
Test Type: static test

Toxicity to daphnia and other : LC50 (Daphnia magna (Water flea)): 0.76 mg/l
aquatic invertebrates : Exposure time: 48 h
Method: OECD Test Guideline 202

LC50: 0.37 mg/l

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

		Exposure time: 96 h Test Type: static test
		EC50: 0.14 mg/l Exposure time: 24 h Test Type: static test
		EC50: 0.072 mg/l Exposure time: 96 h Test Type: static test
Toxicity to algae/aquatic plants	:	IC50 (<i>Pseudokirchneriella subcapitata</i> (algae)): 0.044 mg/l Exposure time: 72 h Method: OECD Test Guideline 201
		NOEC (<i>Pseudokirchneriella subcapitata</i> (algae)): 0.024 mg/l Exposure time: 3 d Method: OECD Test Guideline 201
		IC50 (<i>Skeletonema costatum</i> (marine diatom)): 1.23 mg/l Exposure time: 96 h Method: OECD Test Guideline 201
		IC50: 3.28 mg/l Exposure time: 96 h Method: OECD Test Guideline 201
		NOEC (<i>Dunaliella tertiolecta</i> (marine algae)): 0.01 mg/l Exposure time: 4 d Test Type: static test
		EC50 (<i>Dunaliella tertiolecta</i> (marine algae)): 0.65 mg/l Exposure time: 4 d Test Type: static test
		(<i>Chlorella vulgaris</i> (Fresh water algae)): 1.16 mg/l Exposure time: 72 h Method: OECD Test Guideline 201
		EC50 (<i>Anabaena flos-aquae</i> (cyanobacterium)): 0.3 mg/l Exposure time: 96 h Test Type: static test
		EC50: 0.69 mg/l Exposure time: 3 d Test Type: static test
		EC50 (<i>Phaeodactylum tricornutum</i>): 1.12 mg/l Exposure time: 24 h Test Type: static test
M-Factor (Acute aquatic toxicity)	:	1
Toxicity to fish (Chronic toxicity)	:	NOEC (<i>Oncorhynchus mykiss</i> (rainbow trout)): 0.440 mg/l

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

icity)		Exposure time: 72 d Test Type: flow-through test Remarks: Based on data from similar materials NOEC (Jordanelia floridae (flagfish)): 0.026 mg/l Exposure time: 30 d Method: OECD Test Guideline 210 Remarks: Based on data from similar materials NOEC (Salvelinus fontinalis (Brook trout)): 0.530 mg/l Exposure time: 1,095 d Test Type: flow-through test Remarks: Based on data from similar materials NOEC (Salmo trutta (brown trout)): 0.056 mg/l Exposure time: 116 d Method: OECD Test Guideline 210 Remarks: Based on data from similar materials NOEC (Fish): 0.025 mg/l Exposure time: 27 d Test Type: semi-static test Remarks: Based on data from similar materials NOEC (Pimephales promelas (fathead minnow)): 0.078 mg/l Exposure time: 248 d Test Type: flow-through test Remarks: Based on data from similar materials NOEC (Fish): 0.050 mg/l Exposure time: 155 d Test Type: flow-through test Remarks: Based on data from similar materials
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	LOEC (Daphnia magna (Water flea)): 0.125 mg/l Exposure time: 21 d Method: OECD Test Guideline 211
M-Factor (Chronic aquatic toxicity)	:	10
Toxicity to microorganisms	:	EC50 (activated sludge): > 1,000 mg/l Exposure time: 3 h Method: OECD Test Guideline 209 EC50 (Tetrahymena pyriformis): 7.1 mg/l Exposure time: 24 h Test Type: Growth inhibition
Toxicity to soil dwelling organisms	:	NOEC (Eisenia fetida (earthworms)): 750 mg/kg Exposure time: 21 d
ethanediol:		
Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): > 72,860 mg/l

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : IC50 (Pseudokirchneriella subcapitata (green algae)): 10,940 mg/l
Exposure time: 96 h

Toxicity to fish (Chronic toxicity) : (Menidia peninsulae (tidewater silverside)): 1,500 mg/l
Exposure time: 28 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : (Daphnia magna (Water flea)): 33,911 mg/l
Exposure time: 21 d

Toxicity to microorganisms : (activated sludge): > 1,995 mg/l
Exposure time: 30 min
Method: ISO 8192

1,2-benzisothiazol-3(2H)-one:

Toxicity to fish : LC50 (Cyprinodon variegatus (sheepshead minnow)): 16.7 mg/l
Exposure time: 96 h
Test Type: static test

LC50 (Oncorhynchus mykiss (rainbow trout)): 2.15 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 2.9 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): 0.070 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

NOEC (Pseudokirchneriella subcapitata (green algae)): 0.04 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

M-Factor (Acute aquatic toxicity) : 10

Toxicity to microorganisms : EC50 (activated sludge): 24 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

EC50 (activated sludge): 12.8 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

Persistence and degradability

Components:

zinc oxide:

Biodegradability : Remarks: The methods for determining the biological degradability are not applicable to inorganic substances.

ethanediol:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 90 - 100 %
Exposure time: 10 d
Method: OECD Test Guideline 301A

1,2-benzisothiazol-3(2H)-one:

Biodegradability : Result: rapidly biodegradable
Method: OECD Test Guideline 301C

Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data available

Components:

zinc oxide:

Bioaccumulation : Species: Oncorhynchus mykiss (rainbow trout)
Bioconcentration factor (BCF): 2,060
Exposure time: 14 d
Remarks: Bioaccumulation is unlikely.

ethanediol:

Partition coefficient: n-octanol/water : log Pow: -1.36

1,2-benzisothiazol-3(2H)-one:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Bioconcentration factor (BCF): 6.62
Exposure time: 56 d
Method: OECD Test Guideline 305
Remarks: Substance is not persistent, bioaccumulative, and toxic (PBT).

Partition coefficient: n-octanol/water : log Pow: 0.7 (20 °C)
pH: 7

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0 Revision Date: 26.11.2025 SDS Number: 50001138 Date of last issue: -
Date of first issue: 26.11.2025

log Pow: 0.99 (20 °C)
pH: 5

Mobility in soil

Components:

1,2-benzisothiazol-3(2H)-one:

Distribution among environmental compartments : Koc: 9.33 ml/g, log Koc: 0.97
Method: OECD Test Guideline 121
Remarks: Highly mobile in soils

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

Hygienic standards:

(Allowable concentration in air, water, including fishery waters, soil)

Components	Air	Water	Soil	Data Source
zinc oxide 1314-13-2	MPC - average: 0.05 mg/m ³ (Zinc) Limiting health hazard indicator: resorptive Hazard class: Class 3 - moderately dangerous MPC - average chronic: 0.035 mg/m ³ (Zinc) Limiting health hazard indicator: resorptive Hazard class: Class 3 - moderately dangerous	No data available	No data available	List 1
ethanediol 107-21-1	TSEL: 1 mg/m ³	MPC: 0.25 Milligrams per cubed decimeter Limiting health hazard indicator: sanitary - violation of environmental conditions: changing trophic water	No data available	List 2 List 4 List 5

SAFETY DATA SHEET



THYPAAN ZN

Version 1.0	Revision Date: 26.11.2025	SDS Number: 50001138	Date of last issue: - Date of first issue: 26.11.2025
----------------	------------------------------	-------------------------	--

		bodies fishery; hydrochemical parameters: oxy- gen, nitrogen, phosphorus, pH, impaired self- purification of water bodies of water fishery: BOD5 (bio- chemical oxygen demand for 5 days), the number of saprophytic mi- croflora Hazard class: 4 MPC: 0.5 Milligrams per cubed decimeter Limiting health hazard indicator: sanitary - violation of environmental conditions: chang- ing trophic water bodies fishery; hydrochemical parameters: oxy- gen, nitrogen, phosphorus, pH, impaired self- purification of water bodies of water fishery: BOD5 (bio- chemical oxygen demand for 5 days), the number of saprophytic mi- croflora Hazard class: 3 MAC: 1 mg/l Limiting health hazard indicator: sanitary- toxicological Hazard class: Class 3 - moderately dangerous		
--	--	--	--	--

For explanation of abbreviations see section 16.

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

14. TRANSPORT INFORMATION

ADR

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Zinc oxide)
Class : 9
Packing group : III
Labels : 9
Hazard Identification Number : 90
Tunnel restriction code : (-)
Environmentally hazardous : yes

UNRTDG

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Zinc oxide)
Class : 9
Packing group : III
Labels : 9
Environmentally hazardous : yes

IATA-DGR

UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s.
(Zinc oxide)
Class : 9
Packing group : III
Labels : Miscellaneous
Packing instruction (cargo aircraft) : 964
Packing instruction (passenger aircraft) : 964
Environmentally hazardous : yes

IMDG-Code

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Zinc oxide)
Class : 9
Packing group : III
Labels : 9

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

EmS Code : F-A, S-F
Marine pollutant : yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

The components of this product are reported in the following inventories:

TCSI	: Not in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. Sodium Polyacrylate Homopolymer
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

16. OTHER INFORMATION

Full text of H-Statements

H302	Harmful if swallowed.
H303	May be harmful if swallowed.
H313	May be harmful in contact with skin.
H317	May cause an allergic skin reaction.
H318	Causes serious eye damage.
H373	May cause damage to organs through prolonged or repeated exposure

THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

	if swallowed.
H400	Very toxic to aquatic life.
H402	Harmful to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	: Acute toxicity
Aquatic Acute	: Short-term (acute) aquatic hazard
Aquatic Chronic	: Long-term (chronic) aquatic hazard
Eye Dam.	: Serious eye damage
Skin Sens.	: Skin sensitisation
STOT RE	: Specific target organ toxicity - repeated exposure
List 1	: SanPiN 1.2.3685-21 Table 1.1, Table 1.10, & Table 1.11 Maximum permissible concentration (MPC) in the air of urban and rural settlements
List 2	: SanPiN 1.2.3685-21 Table 1.2, Table 1.12 & Table 1.13 Tentative Safe Exposure Levels (TSEL) in the air of urban and rural settlements
List 4	: SanPiN 1.2.3685-21 Table 3.13, Table 3.15, Table 3.16 & Table 3.17 Maximum permissible concentrations (MPC) of chemicals in the water of drinking systems of centralized, including hot, and non-centralized water supply, water of underground and surface water bodies of domestic drinking and cultural and domestic water use, water of swimming pools, water parks
List 5	: Order of the Russian Federal Fisheries Agency "Standards of maximum permissible concentrations of harmful substances in fishery water bodies"

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); EC_x - Concentration associated with x% response; EL_x - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErC_x - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC₅₀ - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC₅₀ - Lethal Concentration to 50 % of a test population; LD₅₀ - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International

SAFETY DATA SHEET



THYPAAN ZN

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001138	Date of first issue: 26.11.2025

Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

FMC Corporation believes that the information and recommendations contained herein (including data and statements) are accurate as of the date hereof. You can contact FMC Corporation to ensure that this document is the most current available from FMC Corporation. No warranty of fitness for any particular purpose, warranty of merchantability or any other warranty, expressed or implied, is made concerning the information provided herein. The information provided herein relates only to the specified product designated and may not be applicable where such product is used in combination with any other materials or in any process. The user is responsible for determining whether the product is fit for a particular purpose and suitable for the user's conditions and methods of use. Since the conditions and methods of use are beyond the control of FMC Corporation, FMC Corporation expressly disclaims any and all liability as to any results obtained or arising from any use of the products or reliance on such information.

UZ / 6N